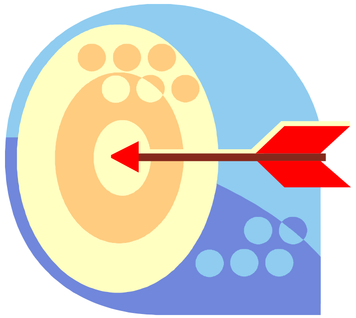
Lesson 03 XPath Basics

Lesson 3 Objectives:



Objectives: At the end of this lesson participants should understand the following:

- Define XPath
- Explain the basic XPath features used to access document content
- Use XPath Visualizer
- Utilize XPath Syntax and Expressions
- Use XPath Expression Builder to build XPath expressions

What is XPath

- XML Path Language (XPath) is a common syntax and semantics for functionality that shared between XSL Transformations and XPointer. The primary purpose of XPath is to address parts of an XML document. In support of this primary purpose, it also provides basic facilities for manipulation of strings, numbers, and booleans. XPath uses a compact, non-XML syntax to facilitate use of XPath within URIs and XML attribute values. XPath operates on the abstract, logical structure of an XML document, rather than its surface syntax.
- XPath expressions can also represent numbers, strings, or Booleans, so XSLT style sheets carry out simple arithmetic for numbering and cross-referencing figures, tables, and equations. String manipulation in XPath allows XSLT perform tasks like making the title of a chapter uppercase in a headline, but mixed case in a reference in the body text.
- **Xpath Version -** As of 2016 there are multiple versions of XPath. 3.0 being the latest version. Sterling B2B Integrator supports XPath 1.0. This is still the most widely adopted of the versions.

Structure of an XML Document

An XML document is a tree made up of nodes. Some nodes contain other nodes. One root node ultimately contains all other nodes. XPath is a language for picking nodes and sets of nodes out of this tree.

There are seven kinds of nodes:

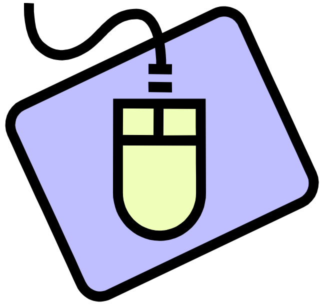
- The root node
- Element nodes
- Text nodes
- Attribute nodes
- Comment nodes
- Processing instruction nodes
- Namespace nodes

XPath Visualizer

Reminder: XPath Visualizer is a third-party software, and is NOT supported by Sterling B2B Integrator Customer Support.

Xpath Visualizer

- XPath Visualizer is a full blown visual XPath Interpreter for the evaluation of any XPath expression and visual presentation of the resulting nodeset or scalar value. The XPath Visualizer's value as an XSLT and XPath learning and authoring tool results from its ability to present the results of any XPath expression in an immediate, appealing, and straightforward visualization.
- A visual interpreter is good aid that can help you learn and test XPath queries and their results. It is not a necessary tool for the operation of Sterling B2B Integrator.
- **Note:** Although this software is old as far as technology is considered it is one of the only XPath interpreters that only supports XPath version 1.0. Newer interpreters can also include newer versions of the XPath standard. This may or may not cause the results during testing to be different than the results in B2Bi.



Exercise: Testing XPath Expressions using XPath Visualizer

Pause this presentation, then follow the instructions to test XPath expressions using XPath Visualizer

XPath Syntax and Expressions

XPath uses path expressions to select nodes or node-sets in an XML document. These path expressions look very much like the expressions you see when you work with a traditional computer file system.

The following types of expressions are used: Your student manual explains these in more detail.

- Locating Nodes
- Wild Cards
- Selective Searching
- Using the “And” operator in searches
- Searching Attributes
- Searching Content

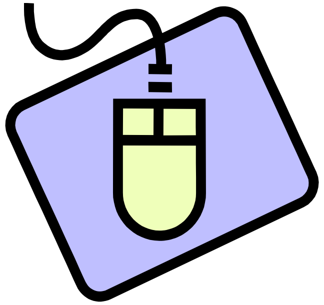
Numeric Expressions - XPath also support mathematical and numeric relational expressions, including equality and Boolean.

XPath Syntax and Expressions (continued)

Strings - Returns the value of an element and an object. This is helpful when the value you extract must be passed as a parameter to another service within a business process.

Substring - Substring allows you to locate and extract part of the content of an element without having to make use of the entire element.

String-length - String-length returns an integer value counting the number of bytes of data exist within an element.

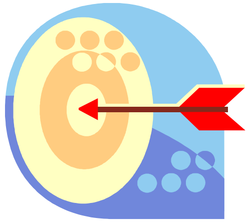


Exercise: Viewing results in XPath Visualizer.

Pause this presentation, then follow the steps to View results in XPath Visualizer.

Sterling B2B Integrator XPath Expression Builder

XPath Expression Builder



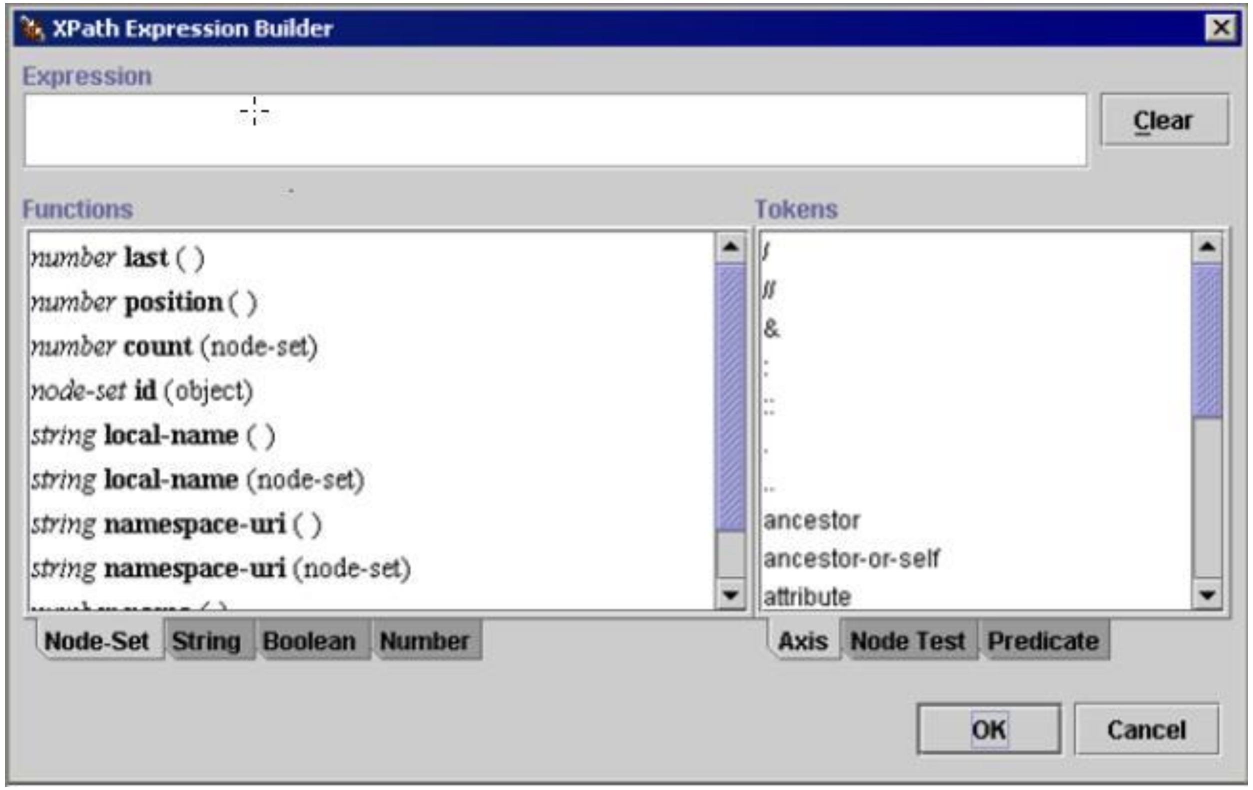
The XPath Expression Builder enables you to:
- Easily create complex rules within a business process—for example, a rule to locate a specific node or source content within an XML document.
- Create and define your own name and value for sending the name-value pairs through business process data.
- Describe the path for sending name-value pairs.

The XPath Expression Builder supports:
- Four expression types: Node-Set, String, Boolean, and Number
- Three token sets: Axis, Node Test, and Predicate

Your student manual provides descriptions of each expression type and token set.

XPath Expression Builder (Continued)

The following figure shows the XPath Expression Builder. The functions for the Node-Set expression type and the Axis token set are visible.



XPath Expression Builder (Continued)

Displaying XPath Expression Builder:
After starting the GPM, you can display the XPath Expression Builder using either of the following features:

- Rule Manager – Assign rules and conditions.
- Service editor – Assign name-value pairs that represent source content and location in business process data.

Knowledge Check

1. "/ProcessData/Order/OrderNumber" is an example of:
a. An absolute path
b. A relative path

2. In XPath, "/text()" and "string" perform the same function
a. True
b. False

3. Which of the following is an example of an indexed search using XPath?
a. "/ProcessData/Order/item/text()"
b. "/ProcessData/Order/item[3]/price"
c. "/ProcessData/Order/*"
d. "/ProcessData/Order/item"

4. Which reserved character is used to denote an attribute in XPath searches?
a. *
b. $
c. @
d. &

Lesson Summary

You should now be familiar with basic XPath features used to access document content.